FRC Sensor CAN · Revision A purchasing BOM

51 order lines · 161 fitted components + 13 shunts · Indicative total **US$49.61**

Dated observations: 2026-09-07. Quantity-one prices are applied to every unit; actual quantity breaks may reduce cost. PCB, stencil, tools, mounting hardware, shipping and tax are excluded. This is not a live quote or reserved inventory. Recheck exact order codes before purchase.

Use DigiKey_import_exact.csv or DigiKey_import_with_spares.csv to import an order. This PDF lists exact assembly quantities. Full ratings, footprint IDs, sources and alternative-review notes are in BOM_grouped.csv. The interactive BOM provides placement guidance.

References	Qty	Value / rating	Manufacturer part number	DigiKey order code	Observed stock	Unit USD	Line USD
F1	1	1.5A FAST	**045101.5MRL** Littelfuse	F2579CT-ND	7,870	2.47	2.47
J1 J2 J3	3	2P 5.00mm	**691137710002** Wurth Elektronik	732-10955-ND	46,145	0.41	1.23
J4 J5 J6 J7	4	3P 5.00mm	**691137710003** Wurth Elektronik	732-10956-ND	5,153	0.66	2.64
Y1	1	8MHz 18pF	**ABM3B-8.000MHZ-B2-T** Abracon	535-9720-1-ND	70,541	0.76	0.76
Q1	1	AO4485	**AO4485** Alpha & Omega Semiconductor	785-1551-1-ND	64,905	1.51	1.51
U2	1	AP2112K-3.3	**AP2112K-3.3TRG1** Diodes Incorporated	AP2112K-3.3TRG1DICT-ND	159,040	0.28	0.28
D102 D103 D202 D203 D302 D303 D402 D403	8	BAT46W	**BAT46W-E3-18** Vishay	BAT46W-E3-18GICT-ND	19,931	0.52	4.16
C25 C26	2	30pF / 50V / 5%	**C0805C300J5GACTU** KEMET	399-C0805C300J5GACTUCT-ND	9,211	0.33	0.66
C101 C201 C301 C401	4	1nF / 50V / 5%	**CC0805JRNPO9BN102** YAGEO	311-1122-1-ND	105,431	0.15	0.60
C10 C23 C42 C7	4	1uF / 50V / 10%	**CC0805KKX7R9BB105** YAGEO	311-1886-1-ND	112,870	0.32	1.28
C28 C29	2	10nF / 50V / 10%	**CC0805KRX7R9BB103** YAGEO	311-1136-1-ND	360,198	0.11	0.22
C11 C20 C21 C22 C27 C3 C40 C41 C50 C6 C9	11	100nF / 50V / 10%	**CC0805KRX7R9BB104** YAGEO	311-1140-1-ND	1,921,549	0.11	1.21

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C14	1	33nF / 50V / 10%	**CC0805KRX7R9BB333** YAGEO	311-1374-1-ND	24,641	0.14	0.14
C12 C13 C4 C5	4	22uF / 25V / 10%	**CL31A226KAHNNNE** Samsung Electro-Mechanics	1276-3047-1-ND	426,249	0.48	1.92
C2	1	10uF / 50V / 10%	**CL32B106KBJNNWE** Samsung Electro-Mechanics	1276-3388-1-ND	149,003	0.62	0.62
C1	1	47uF / 35V / 20%	**EEE-FK1V470P** Panasonic Industry	PCE3842CT-ND	1,183	0.55	0.55
R107 R207 R307 R407	4	17.8k / 1% / 0.125 W	**ERJ-6ENF1782V** Panasonic Industry	P17.8KCCT-ND	33,381	0.12	0.48
D20	1	ESD2CAN24-Q1	**ESD2CAN24DBZRQ1** Texas Instruments	296-ESD2CAN24DBZRQ1CT-ND	255,809	0.87	0.87
U7	1	LM2901BIDR	**LM2901BIDR** Texas Instruments	296-LM2901BIDRCT-ND	3,349	0.29	0.29
U1	1	LMR51420XDDCR	**LMR51420XDDCR** Texas Instruments	296-LMR51420XDDCRCT-ND	14,239	1.32	1.32
D10 D12 D13 D14 D15	5	GREEN	**LTST-C170KGKT** Lite-On	160-1414-1-ND	498,450	0.19	0.95
D11	1	RED	**LTST-C170KRKT** Lite-On	160-1415-1-ND	489,805	0.18	0.18
JP103 JP203 JP303 JP403	4	SELECT	**M20-9980345** Harwin	952-2120-ND	14,910	0.29	1.16
J8	1	DEBUG	**M20-9980545** Harwin	952-2380-ND	11,669	0.45	0.45
JP1 JP102 JP2 JP202 JP302 JP402	6	SELECT	**M20-9990245** Harwin	952-2261-ND	66,017	0.12	0.72
JP101 JP201 JP301 JP401	4	SELECT	**M20-9990345** Harwin	952-2263-ND	75,301	0.15	0.60
D3	1	12V ZENER	**MMSZ12T1G** onsemi	MMSZ12T1GOSCT-ND	34,429	0.17	0.17
Separate shunts	13	2.54mm shorting shunt	**QPC02SXGN-RC** Sullins Connector Solutions	S9337-ND	2,159,145	0.10	1.30

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R106 R12 R206 R23 R306 R406 R7	7	100k / 1% / 0.125 W	**RC0805FR-07100KL** YAGEO	311-100KCRCT-ND	1,152,791	0.11	0.77
R105 R11 R14 R15 R205 R21 R22 R24 R25 R26 R3 R305 R40 R405 R41 R6 R9	17	10k / 1% / 0.125 W	**RC0805FR-0710KL** YAGEO	311-10.0KCRCT-ND	2,321,007	0.10	1.70
R101 R102 R201 R202 R301 R302 R401 R402	8	110k / 1% / 0.125 W	**RC0805FR-07110KL** YAGEO	311-110KCRCT-ND	5,943	0.10	0.80
R13 R30 R31 R32 R33 R34 R35 R4	8	1k / 1% / 0.125 W	**RC0805FR-071KL** YAGEO	311-1.00KCRCT-ND	894,200	0.10	0.80
R103 R203 R303 R403	4	20k / 1% / 0.125 W	**RC0805FR-0720KL** YAGEO	311-20.0KCRCT-ND	27,443	0.10	0.40
R5	1	3.9k / 1% / 0.125 W	**RC0805FR-073K9L** YAGEO	311-3.90KCRCT-ND	131,203	0.10	0.10
R110 R111 R210 R211 R310 R311 R410 R411 R8	9	4.7k / 1% / 0.125 W	**RC0805FR-074K7L** YAGEO	311-4.70KCRCT-ND	564,468	0.10	0.90
R10	1	59k / 1% / 0.125 W	**RC0805FR-0759KL** YAGEO	311-59.0KCRCT-ND	29,786	0.10	0.10
R108 R208 R308 R408	4	7.15k / 1% / 0.125 W	**RC0805FR-077K15L** YAGEO	311-7.15KCRCT-ND	48,097	0.10	0.40
R104 R204 R304 R404	4	910k / 1% / 0.125 W	**RC0805FR-07910KL** YAGEO	YAG3336CT-ND	58,915	0.10	0.40
R20	1	0 / jumper / 0.125 W	**RC0805JR-070RL** YAGEO	311-0.0ARCT-ND	1,733,368	0.10	0.10
R42	1	120 / 1% / 0.25 W	**RC1206FR-07120RL** YAGEO	311-120FRCT-ND	126,570	0.10	0.10
R109 R209 R309 R409	4	4.7k / 1% / 0.75 W	**RC2010FK-074K7L** YAGEO	13-RC2010FK-074K7LCT-ND	7,566	0.23	0.92
R1	1	101k / 0.1% / 0.125 W	**RT0805BRD07101KL** YAGEO	13-RT0805BRD07101KLCT-ND	4,939	0.10	0.10
R2	1	13.7k / 0.1% / 0.125 W	**RT0805BRD0713K7L** YAGEO	YAG4829CT-ND	4,402	0.10	0.10
D101 D201 D301 D401	4	SMAJ30A	**SMAJ30A** Littelfuse	SMAJ30ALFCT-ND	104,069	0.50	2.00

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D2	1	SMBJ18A	**SMBJ18A-E3/52** Vishay	SMBJ18A-E3/52GICT-ND	38,516	0.51	0.51
L1	1	10uH	**SRP7028A-100M** Bourns	SRP7028A-100MCT-ND	9,366	1.10	1.10
U5	1	STM32F042K6T6	**STM32F042K6T6** STMicroelectronics	497-14647-ND	3,072	4.34	4.34
U6	1	TCAN3413DR	**TCAN3413DR** Texas Instruments	296-TCAN3413DRCT-ND	5,237	1.87	1.87
C24 C8	2	4.7uF / 25V / 10%	**TMK212AB7475KG-T** Taiyo Yuden	587-2990-1-ND	718,393	0.26	0.52
U3	1	TPS1H200A-Q1	**TPS1H200AQDGNRQ1** Texas Instruments	296-50416-1-ND	25,656	1.78	1.78
U4	1	TPS2553DBVR-1	**TPS2553DBVR-1** Texas Instruments	296-39380-1-ND	5,463	1.06	1.06